# RHEL 6

## SAN hosts and cloud clients

NetApp
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# RHEL 6

## Use Red Hat Enterprise Linux 6.10 with ONTAP

You can use the ONTAP SAN host configuration settings to configure Red Hat Enterprise Linux 6.10 with ONTAP as the target.

### Install the Linux Unified Host Utilities

The NetApp Linux Unified Host Utilities software package is available on the NetApp Support Site in a 32-bit and 64-bit .rpm file. If you do not know which file is right for your configuration, use the NetApp Interoperability Matrix Tool to verify which one you need.

NetApp strongly recommends installing the Linux Unified Host Utilities, but it is not mandatory. The utilities do not change any settings on your Linux host. The utilities improve management and assist NetApp customer support in gathering information about your configuration.

#### What you'll need

If you have a version of Linux Unified Host Utilities currently installed you should upgrade it or, you should remove it and use the following steps to install the latest version.

1. Download the 32-bit or 64-bit Linux Unified Host Utilities software package from the NetApp Support Site Site to your host.
2. Use the following command to install the software package:

   ```
   rpm -ivh netapp_linux_unified_host_utilities-7-1.x86_64
   ```

> You can use the configuration settings provided in this document to configure cloud clients connected to Cloud Volumes ONTAP and Amazon FSx for ONTAP.

### SAN Toolkit

The toolkit is installed automatically when you install the NetApp Host Utilities package. This kit provides the `sanlun` utility, which helps you manage LUNs and HBAs. The `sanlun` command returns information about the LUNs mapped to your host, multipathing, and information necessary to create initiator groups.

#### Example

In the following example, the `sanlun lun show` command returns LUN information.

```
# sanlun lun show all
```

Example output:

```
controller(7mode/E-Series)/            device     host                lun
vserver(cDOT/FlashRay)   lun-pathname  filename   adapter  protocol  size
Product
-------------------------------------------------------------------------
data_vserver          /vol/vol1/lun1   /dev/sdb   host16   FCP
120.0g  cDOT
data_vserver          /vol/vol1/lun1   /dev/sdc   host15   FCP
120.0g  cDOT
data_vserver          /vol/vol2/lun2   /dev/sdd   host16   FCP
120.0g  cDOT
data_vserver          /vol/vol2/lun2   /dev/sde   host15   FCP
120.0g  cDOT
```

## SAN Booting

### What you'll need

If you decide to use SAN booting, it must be supported by your configuration. You can use the NetApp Interoperability Matrix Tool to verify that your OS, HBA, HBA firmware and the HBA boot BIOS, and ONTAP version are supported.

### Steps

1. Map the SAN boot LUN to the host.
2. Verify that multiple paths are available.

   Multiple paths become available after the host OS is up and running on the paths.

3. Enable SAN booting in the server BIOS for the ports to which the SAN boot LUN is mapped.

   For information on how to enable the HBA BIOS, see your vendor-specific documentation.

4. Reboot the host to verify that the boot is successful.

## Multipathing

For Red Hat Enterprise Linux (RHEL) 6.10 the /etc/multipath.conf file must exist, but you do not need to make specific changes to the file. RHEL 6.10 is compiled with all settings required to recognize and correctly manage ONTAP LUNs.
To Enable ALUA Handler, perform the following steps:

### Steps

1. Create a backup of the initrd-image.
2. Append the following parameter value to the kernel for ALUA and non-ALUA to work:
   `rdloaddriver=scsi_dh_alua`

**Example**

```
kernel /vmlinuz-2.6.32-358.6.1.el6.x86_64 ro root=/dev/mapper/
vg_ibmx355021082-lv_root rd_NO_LUKS rd_LVM_LV=vg_ibmx355021082/ lv_root
LANG=en_US.UTF-8 rd_LVM_LV=vg_ibmx355021082/lv_swap rd_NO_MD
SYSFONT=latarcyrheb-sun16 crashkernel=auto KEYBOARDTYPE=pc KEYTABLE=us
rd_NO_DM rhgb quiet rdloaddriver=scsi_dh_alua
```

3. Use the `mkinitrd` command to recreate the initrd-image.
   RHEL 6x and later versions use either:
   The command: `mkinitrd -f /boot/ initrd-"uname -r".img uname -r`
   Or
   The command: `dracut -f`
4. Reboot the host.
5. Verify the output of the `cat /proc/cmdline` command to ensure that the setting is complete.

You can use the `multipath -ll` command to verify the settings for your ONTAP LUNs.
The following sections provide sample multipath output for a LUN mapped to ASA and non-ASA personas.

### All SAN Array configurations

In All SAN Array (ASA) configurations, all paths to a given LUN are active and optimized. This improves performance by serving I/O operations through all paths at the same time.

**Example**

The following example displays the correct output for an ONTAP LUN:

```
# multipath -ll
3600a0980383034466b2b4a3775474859 dm-3 NETAPP,LUN C-Mode
size=20G features='4 queue_if_no_path pg_init_retries 50
retain_attached_hw_handle' hwhandler='1 alua' wp=rw
|-+- policy='round-robin 0' prio=50 status=active
|- 1:0:8:1  sdb 8:16 active ready running
|- 2:0:8:1  sdd 8:48 active ready running
|- 1:0:9:1  sdc 8:32 active ready running
|- 2:0:9:1  sde 8:64 active ready running
```

> Do not use an excessive number of paths to a single LUN. No more than four paths should be required. More than eight paths might cause path issues during storage failures.

### Non-ASA configurations

For non-ASA configurations, there should be two groups of paths with different priorities. The paths with the higher priorities are Active/Optimized, meaning they are serviced by the controller where the aggregate is located. The paths with the lower priorities are active but are non-optimized because they are served from a different controller. The non-optimized paths are only used when no optimized paths are available.

**Example**

The following example displays the correct output for an ONTAP LUN with two Active/Optimized paths and two Active/Non-Optimized paths:

```
# multipath -ll
3600a0980383034466b2b4a3775474859 dm-3 NETAPP,LUN C-Mode
size=20G features='4 queue_if_no_path pg_init_retries 50
retain_attached_hw_handle' hwhandler='1 alua' wp=rw
|-+- policy='round-robin 0' prio=50 status=active
| |- 1:0:8:1  sdb 8:16 active ready running
| `- 2:0:8:1  sdd 8:48 active ready running
`-+- policy='round-robin 0' prio=10 status=enabled
  |- 1:0:9:1  sdc 8:32 active ready running
  `- 2:0:9:1  sde 8:64 active ready running
```

Do not use an excessive number of paths to a single LUN. No more than four paths should be required. More than eight paths might cause path issues during storage failures.

## Recommended Settings

The RHEL 6.10 OS is compiled to recognize ONTAP LUNs and automatically set all configuration parameters correctly for both ASA and non-ASA configuration.

The `multipath.conf` file must exist for the multipath daemon to start, but you can create an empty, zero-byte file by using the following command:

```
touch /etc/multipath.conf
```

The first time you create this file, you might need to enable and start the multipath services:

```
# systemctl enable multipathd
# systemctl start multipathd
```

- There is no requirement to add anything directly to the `multipath.conf` file, unless you have devices that you do not want to be managed by multipath or you have existing settings that override defaults.
- To exclude unwanted devices, add the following syntax to the `multipath.conf` file .

  ```
  blacklist {
          wwid <DevId>
          devnode "^(ram|raw|loop|fd|md|dm-|sr|scd|st)[0-9]*"
          devnode "^hd[a-z]"
          devnode "^cciss.*"
  }
  ```

  Replace the `<DevId>` with the `WWID` string of the device you want to exclude.

**Example**

In this example, we are going to determine the WWID of a device and add to the `multipath.conf` file.

**Steps**

1. Run the following command to determine the WWID:

   ```
   # /lib/udev/scsi_id -gud /dev/sda
   360030057024d0730239134810c0cb833
   ```

   `sda` is the local SCSI disk that we need to add it to the blacklist.

2. Add the `WWID` to the blacklist stanza in `/etc/multipath.conf`:

   ```
   blacklist {
        wwid   360030057024d0730239134810c0cb833
        devnode "^(ram|raw|loop|fd|md|dm-|sr|scd|st)[0-9]*"
        devnode "^hd[a-z]"
        devnode "^cciss.*"
   }
   ```

You should always check your `/etc/multipath.conf` file for legacy settings, especially in the defaults section, that might be overriding the default settings.

The following table demonstrates the critical `multipathd` parameters for ONTAP LUNs and the required values. If a host is connected to LUNs from other vendors and any of these parameters are overridden, they will need to be corrected by later stanzas in the `multipath.conf` file that apply specifically to ONTAP LUNs. If this is not done, the ONTAP LUNs might not work as expected. These defaults should only be overridden in consultation with NetApp and/or an OS vendor and only when the impact is fully understood.

| Parameter | Setting |
|---|---|
| detect_prio | yes |
| dev_loss_tmo | "infinity" |
| failback | immediate |
| fast_io_fail_tmo | 5 |
| features | "3 queue_if_no_path pg_init_retries 50" |
| flush_on_last_del | "yes" |
| hardware_handler | "0" |
| no_path_retry | queue |
| path_checker | "tur" |
| path_grouping_policy | "group_by_prio" |
| path_selector | "round-robin 0" |

| Parameter | Setting |
| --- | --- |
| polling_interval | 5 |
| prio | "ontap" |
| product | LUN.* |
| retain_attached_hw_handler | yes |
| rr_weight | "uniform" |
| user_friendly_names | no |
| vendor | NETAPP |

#### Example

The following example shows how to correct an overridden default. In this case, the `multipath.conf` file defines values for `path_checker` and `no_path_retry` that are not compatible with ONTAP LUNs. If they cannot be removed because of other SAN arrays still attached to the host, these parameters can be corrected specifically for ONTAP LUNs with a device stanza.

```
defaults {
   path_checker      readsector0
   no_path_retry      fail
}

devices {
   device {
      vendor         "NETAPP  "
      product         "LUN.*"
      no_path_retry     queue
      path_checker      tur
   }
}
```

#### KVM settings

You can use the recommended settings to configure Kernel-based Virtual Machine (KVM) as well. There are no changes required to configure KVM as the LUN is mapped to the hypervisor.

### Known issues

There are no known issues for the RHEL 6.10 with ONTAP release.

# Use Red Hat Enterprise Linux 6.9 with ONTAP

You can use the ONTAP SAN host configuration settings to configure Red Hat Enterprise Linux 6.9 with ONTAP as the target.

## Install the Linux Unified Host Utilities

The NetApp Linux Unified Host Utilities software package is available on the NetApp Support Site in a 32-bit and 64-bit .rpm file. If you do not know which file is right for your configuration, use the NetApp Interoperability Matrix Tool to verify which one you need.

NetApp strongly recommends installing the Linux Unified Host Utilities, but it is not mandatory. The utilities do not change any settings on your Linux host. The utilities improve management and assist NetApp customer support in gathering information about your configuration.

#### What you'll need

If you have a version of Linux Unified Host Utilities currently installed you should upgrade it or, you should remove it and use the following steps to install the latest version.

1. Download the 32-bit or 64-bit Linux Unified Host Utilities software package from the NetApp Support Site Site to your host.
2. Use the following command to install the software package:

   ```
   rpm -ivh netapp_linux_unified_host_utilities-7-1.x86_64
   ```

> You can use the configuration settings provided in this document to configure cloud clients connected to Cloud Volumes ONTAP and Amazon FSx for ONTAP.

## SAN Toolkit

The toolkit is installed automatically when you install the NetApp Host Utilities package. This kit provides the `sanlun` utility, which helps you manage LUNs and HBAs. The `sanlun` command returns information about the LUNs mapped to your host, multipathing, and information necessary to create initiator groups.

#### Example

In the following example, the `sanlun lun show` command returns LUN information.

```
# sanlun lun show all
```

Example output:

```
controller(7mode/E-Series)/            device     host               lun
vserver(cDOT/FlashRay)   lun-pathname  filename   adapter  protocol  size
Product
-------------------------------------------------------------------------
data_vserver          /vol/vol1/lun1   /dev/sdb   host16   FCP
120.0g  cDOT
data_vserver          /vol/vol1/lun1   /dev/sdc   host15   FCP
120.0g  cDOT
data_vserver          /vol/vol2/lun2   /dev/sdd   host16   FCP
120.0g  cDOT
data_vserver          /vol/vol2/lun2   /dev/sde   host15   FCP
120.0g  cDOT
```

## SAN Booting

### What you'll need

If you decide to use SAN booting, it must be supported by your configuration. You can use the NetApp Interoperability Matrix Tool to verify that your OS, HBA, HBA firmware and the HBA boot BIOS, and ONTAP version are supported.

### Steps

1. Map the SAN boot LUN to the host.
2. Verify that multiple paths are available.

   Multiple paths become available after the host OS is up and running on the paths.

3. Enable SAN booting in the server BIOS for the ports to which the SAN boot LUN is mapped.

   For information on how to enable the HBA BIOS, see your vendor-specific documentation.

4. Reboot the host to verify that the boot is successful.

## Multipathing

For Red Hat Enterprise Linux (RHEL) 6.9 the /etc/multipath.conf file must exist, but you do not need to make specific changes to the file. RHEL 6.9 is compiled with all settings required to recognize and correctly manage ONTAP LUNs.
To Enable ALUA Handler, perform the following steps:

### Steps

1. Create a backup of the initrd-image.
2. Append the following parameter value to the kernel for ALUA and non-ALUA to work:
   `rdloaddriver=scsi_dh_alua`

**Example**

```
kernel /vmlinuz-2.6.32-358.6.1.el6.x86_64 ro root=/dev/mapper/
vg_ibmx355021082-lv_root rd_NO_LUKS rd_LVM_LV=vg_ibmx355021082/ lv_root
LANG=en_US.UTF-8 rd_LVM_LV=vg_ibmx355021082/lv_swap rd_NO_MD
SYSFONT=latarcyrheb-sun16 crashkernel=auto KEYBOARDTYPE=pc KEYTABLE=us
rd_NO_DM rhgb quiet rdloaddriver=scsi_dh_alua
```

3. Use the `mkinitrd` command to recreate the initrd-image.
   RHEL 6x and later versions use either:
   The command: `mkinitrd -f /boot/ initrd-"uname -r".img uname -r`
   Or
   The command: `dracut -f`
4. Reboot the host.
5. Verify the output of the `cat /proc/cmdline` command to ensure that the setting is complete.

You can use the `multipath -ll` command to verify the settings for your ONTAP LUNs.
The following sections provide sample multipath output for a LUN mapped to ASA and non-ASA personas.

### All SAN Array configurations

In All SAN Array (ASA) configurations, all paths to a given LUN are active and optimized. This improves performance by serving I/O operations through all paths at the same time.

**Example**

The following example displays the correct output for an ONTAP LUN:

```
# multipath -ll
3600a0980383034466b2b4a3775474859 dm-3 NETAPP,LUN C-Mode
size=20G features='4 queue_if_no_path pg_init_retries 50
retain_attached_hw_handle' hwhandler='1 alua' wp=rw
|-+- policy='round-robin 0' prio=50 status=active
|- 1:0:8:1  sdb 8:16 active ready running
|- 2:0:8:1  sdd 8:48 active ready running
|- 1:0:9:1  sdc 8:32 active ready running
|- 2:0:9:1  sde 8:64 active ready running
```

> Do not use an excessive number of paths to a single LUN. No more than four paths should be required. More than eight paths might cause path issues during storage failures.

### Non-ASA configurations

For non-ASA configurations, there should be two groups of paths with different priorities. The paths with the higher priorities are Active/Optimized, meaning they are serviced by the controller where the aggregate is located. The paths with the lower priorities are active but are non-optimized because they are served from a different controller. The non-optimized paths are only used when no optimized paths are available.

**Example**

The following example displays the correct output for an ONTAP LUN with two Active/Optimized paths and two Active/Non-Optimized paths:

```
# multipath -ll
3600a0980383034466b2b4a3775474859 dm-3 NETAPP,LUN C-Mode
size=20G features='4 queue_if_no_path pg_init_retries 50
retain_attached_hw_handle' hwhandler='1 alua' wp=rw
|-+- policy='round-robin 0' prio=50 status=active
| |- 1:0:8:1  sdb 8:16 active ready running
| `- 2:0:8:1  sdd 8:48 active ready running
`-+- policy='round-robin 0' prio=10 status=enabled
  |- 1:0:9:1  sdc 8:32 active ready running
  `- 2:0:9:1  sde 8:64 active ready running
```

Do not use an excessive number of paths to a single LUN. No more than four paths should be required. More than eight paths might cause path issues during storage failures.

## Recommended Settings

The RHEL 6.9 OS is compiled to recognize ONTAP LUNs and automatically set all configuration parameters correctly for both ASA and non-ASA configuration.

The `multipath.conf` file must exist for the multipath daemon to start, but you can create an empty, zero-byte file by using the following command:

```
touch /etc/multipath.conf
```

The first time you create this file, you might need to enable and start the multipath services:

```
# systemctl enable multipathd
# systemctl start multipathd
```

- There is no requirement to add anything directly to the `multipath.conf` file, unless you have devices that you do not want to be managed by multipath or you have existing settings that override defaults.
- To exclude unwanted devices, add the following syntax to the `multipath.conf` file .

  ```
  blacklist {
          wwid <DevId>
          devnode "^(ram|raw|loop|fd|md|dm-|sr|scd|st)[0-9]*"
          devnode "^hd[a-z]"
          devnode "^cciss.*"
  }
  ```

  Replace the `<DevId>` with the `WWID` string of the device you want to exclude.

**Example**

In this example, we are going to determine the WWID of a device and add to the `multipath.conf` file.

**Steps**

1. Run the following command to determine the WWID:

```
# /lib/udev/scsi_id -gud /dev/sda
360030057024d0730239134810c0cb833
```

   `sda` is the local SCSI disk that we need to add it to the blacklist.

2. Add the `WWID` to the blacklist stanza in `/etc/multipath.conf`:

```
blacklist {
     wwid   360030057024d0730239134810c0cb833
     devnode "^(ram|raw|loop|fd|md|dm-|sr|scd|st)[0-9]*"
     devnode "^hd[a-z]"
     devnode "^cciss.*"
}
```

You should always check your `/etc/multipath.conf` file for legacy settings, especially in the defaults section, that might be overriding the default settings.

The following table demonstrates the critical `multipathd` parameters for ONTAP LUNs and the required values. If a host is connected to LUNs from other vendors and any of these parameters are overridden, they will need to be corrected by later stanzas in the `multipath.conf` file that apply specifically to ONTAP LUNs. If this is not done, the ONTAP LUNs might not work as expected. These defaults should only be overridden in consultation with NetApp and/or an OS vendor and only when the impact is fully understood.

| Parameter | Setting |
|---|---|
| detect_prio | yes |
| dev_loss_tmo | "infinity" |
| failback | immediate |
| fast_io_fail_tmo | 5 |
| features | "3 queue_if_no_path pg_init_retries 50" |
| flush_on_last_del | "yes" |
| hardware_handler | "0" |
| no_path_retry | queue |
| path_checker | "tur" |
| path_grouping_policy | "group_by_prio" |
| path_selector | "round-robin 0" |

| Parameter | Setting |
| --- | --- |
| polling_interval | 5 |
| prio | "ontap" |
| product | LUN.* |
| retain_attached_hw_handler | yes |
| rr_weight | "uniform" |
| user_friendly_names | no |
| vendor | NETAPP |

#### Example

The following example shows how to correct an overridden default. In this case, the `multipath.conf` file defines values for `path_checker` and `no_path_retry` that are not compatible with ONTAP LUNs. If they cannot be removed because of other SAN arrays still attached to the host, these parameters can be corrected specifically for ONTAP LUNs with a device stanza.

```
defaults {
   path_checker      readsector0
   no_path_retry      fail
}

devices {
   device {
      vendor         "NETAPP  "
      product         "LUN.*"
      no_path_retry     queue
      path_checker      tur
   }
}
```

#### KVM settings

You can use the recommended settings to configure Kernel-based Virtual Machine (KVM) as well. There are no changes required to configure KVM as the LUN is mapped to the hypervisor.

## Known issues

The RHEL 6.9 with ONTAP release has the following known issues:

| NetApp Bug ID | Title | Description | Bugzilla ID |
| --- | --- | --- | --- |
| 1067272 | Remote port status on EMULEX LPe32002 host might be in 'Blocked' state during storage failover operations | During storage failover operations, certain remote port status on RHEL 6.9 host with LPe32002 adapter might get into 'Blocked' state. Because the logical interfaces go down when a storage node is down, the remote port sets the storage node status to "Blocked" state. However, when the storage node comes back to optimal state, the logical interfaces also comes up and the remote port state is expected to be 'Online'. But, on certain occasion the remote port continues to be in 'Blocked' state. This state manifests as 'failed faulty' to LUNS at multipath layer. | 427496 |

| NetApp Bug ID | Title | Description | Bugzilla ID |
| --- | --- | --- | --- |
| 1076584 | Firmware dumps occur on Red Hat Enterprise Linux 6.9 QLogic QE8362 HBA during storage failover operations | Firmware dumps can occur during storage failover operations on Red Hat Enterprise Linux (RHEL) 6.9 hosts with QLogic QLE8362 host bus adapters (HBA), firmware dumps are observed occasionally. The firmware dumps might manifest as an I/O outage on the host that can last as long as 1200 seconds. After the adapter completes dumping the firmware cores, the I/O operation resumes normally. No further recovery procedure is required on the host. To indicate the firmware dump, the following message is displayed in /var/log/ message file: kernel: qla2xxx [0000:0c:00.3]-d001:3: Firmware dump saved to temp buffer (3/ffffc90018b01000), dump status flags (0x3f) | 1438711 |

# Use Red Hat Enterprise Linux 6.8 with ONTAP

You can use the ONTAP SAN host configuration settings to configure Red Hat Enterprise Linux 6.8 with ONTAP as the target.

## Install the Linux Unified Host Utilities

The NetApp Linux Unified Host Utilities software package is available on the NetApp Support Site in a 32-bit and 64-bit .rpm file. If you do not know which file is right for your configuration, use the NetApp Interoperability Matrix Tool to verify which one you need.

NetApp strongly recommends installing the Linux Unified Host Utilities, but it is not mandatory. The utilities do not change any settings on your Linux host. The utilities improve management and assist NetApp customer support in gathering information about your configuration.

#### What you'll need

If you have a version of Linux Unified Host Utilities currently installed you should upgrade it or, you should remove it and use the following steps to install the latest version.

1. Download the 32-bit or 64-bit Linux Unified Host Utilities software package from the NetApp Support Site Site to your host.

2. Use the following command to install the software package:

```
rpm -ivh netapp_linux_unified_host_utilities-7-1.x86_64
```

> You can use the configuration settings provided in this document to configure cloud clients connected to Cloud Volumes ONTAP and Amazon FSx for ONTAP.

## SAN Toolkit

The toolkit is installed automatically when you install the NetApp Host Utilities package. This kit provides the `sanlun` utility, which helps you manage LUNs and HBAs. The `sanlun` command returns information about the LUNs mapped to your host, multipathing, and information necessary to create initiator groups.

### Example

In the following example, the `sanlun lun show` command returns LUN information.

```
# sanlun lun show all
```

Example output:

```
controller(7mode/E-Series)/            device     host               lun
vserver(cDOT/FlashRay)   lun-pathname  filename   adapter  protocol  size
Product
------------------------------------------------------------------------
data_vserver          /vol/vol1/lun1   /dev/sdb   host16   FCP
120.0g  cDOT
data_vserver          /vol/vol1/lun1   /dev/sdc   host15   FCP
120.0g  cDOT
data_vserver          /vol/vol2/lun2   /dev/sdd   host16   FCP
120.0g  cDOT
data_vserver          /vol/vol2/lun2   /dev/sde   host15   FCP
120.0g  cDOT
```

## SAN Booting

### What you'll need

If you decide to use SAN booting, it must be supported by your configuration. You can use the NetApp Interoperability Matrix Tool to verify that your OS, HBA, HBA firmware and the HBA boot BIOS, and ONTAP version are supported.

### Steps

1. Map the SAN boot LUN to the host.
2. Verify that multiple paths are available.

> Multiple paths become available after the host OS is up and running on the paths.

3. Enable SAN booting in the server BIOS for the ports to which the SAN boot LUN is mapped.

   For information on how to enable the HBA BIOS, see your vendor-specific documentation.

4. Reboot the host to verify that the boot is successful.

## Multipathing

For Red Hat Enterprise Linux (RHEL) 6.8 the /etc/multipath.conf file must exist, but you do not need to make specific changes to the file. RHEL 6.8 is compiled with all settings required to recognize and correctly manage ONTAP LUNs.
To Enable ALUA Handler, perform the following steps:

**Steps**

1. Create a backup of the initrd-image.
2. Append the following parameter value to the kernel for ALUA and non-ALUA to work:
   `rdloaddriver=scsi_dh_alua`

   **Example**

   ```
   kernel /vmlinuz-2.6.32-358.6.1.el6.x86_64 ro root=/dev/mapper/
   vg_ibmx355021082-lv_root rd_NO_LUKS rd_LVM_LV=vg_ibmx355021082/ lv_root
   LANG=en_US.UTF-8 rd_LVM_LV=vg_ibmx355021082/lv_swap rd_NO_MD
   SYSFONT=latarcyrheb-sun16 crashkernel=auto KEYBOARDTYPE=pc KEYTABLE=us
   rd_NO_DM rhgb quiet rdloaddriver=scsi_dh_alua
   ```

3. Use the `mkinitrd` command to recreate the initrd-image.
   RHEL 6x and later versions use either:
   The command: `mkinitrd -f /boot/ initrd-"uname -r".img uname -r`
   Or
   The command: `dracut -f`
4. Reboot the host.
5. Verify the output of the `cat /proc/cmdline` command to ensure that the setting is complete.

You can use the `multipath -ll` command to verify the settings for your ONTAP LUNs.
The following sections provide sample multipath output for a LUN mapped to ASA and non-ASA personas.

### All SAN Array configurations

In All SAN Array (ASA) configurations, all paths to a given LUN are active and optimized. This improves performance by serving I/O operations through all paths at the same time.

**Example**

The following example displays the correct output for an ONTAP LUN:

```
# multipath -ll
3600a0980383034466b2b4a3775474859 dm-3 NETAPP,LUN C-Mode
size=20G features='4 queue_if_no_path pg_init_retries 50
retain_attached_hw_handle' hwhandler='1 alua' wp=rw
|-+- policy='round-robin 0' prio=50 status=active
|- 1:0:8:1  sdb 8:16 active ready running
|- 2:0:8:1  sdd 8:48 active ready running
|- 1:0:9:1  sdc 8:32 active ready running
|- 2:0:9:1  sde 8:64 active ready running
```

Do not use an excessive number of paths to a single LUN. No more than four paths should be required. More than eight paths might cause path issues during storage failures.

#### Non-ASA configurations

For non-ASA configurations, there should be two groups of paths with different priorities. The paths with the higher priorities are Active/Optimized, meaning they are serviced by the controller where the aggregate is located. The paths with the lower priorities are active but are non-optimized because they are served from a different controller. The non-optimized paths are only used when no optimized paths are available.

**Example**

The following example displays the correct output for an ONTAP LUN with two Active/Optimized paths and two Active/Non-Optimized paths:

```
# multipath -ll
3600a0980383034466b2b4a3775474859 dm-3 NETAPP,LUN C-Mode
size=20G features='4 queue_if_no_path pg_init_retries 50
retain_attached_hw_handle' hwhandler='1 alua' wp=rw
|-+- policy='round-robin 0' prio=50 status=active
| |- 1:0:8:1  sdb 8:16 active ready running
| `- 2:0:8:1  sdd 8:48 active ready running
`-+- policy='round-robin 0' prio=10 status=enabled
  |- 1:0:9:1  sdc 8:32 active ready running
  `- 2:0:9:1  sde 8:64 active ready running
```

Do not use an excessive number of paths to a single LUN. No more than four paths should be required. More than eight paths might cause path issues during storage failures.

## Recommended Settings

The RHEL 6.8 OS is compiled to recognize ONTAP LUNs and automatically set all configuration parameters correctly for both ASA and non-ASA configuration.

The `multipath.conf` file must exist for the multipath daemon to start, but you can create an empty, zero-byte file by using the following command:

```
touch /etc/multipath.conf
```

The first time you create this file, you might need to enable and start the multipath services:

```
# systemctl enable multipathd
# systemctl start multipathd
```

- There is no requirement to add anything directly to the `multipath.conf` file, unless you have devices that you do not want to be managed by multipath or you have existing settings that override defaults.
- To exclude unwanted devices, add the following syntax to the `multipath.conf` file .

  ```
  blacklist {
          wwid <DevId>
          devnode "^(ram|raw|loop|fd|md|dm-|sr|scd|st)[0-9]*"
          devnode "^hd[a-z]"
          devnode "^cciss.*"
  }
  ```

  Replace the `<DevId>` with the `WWID` string of the device you want to exclude.

  **Example**

  In this example, we are going to determine the WWID of a device and add to the `multipath.conf` file.

  **Steps**

  1. Run the following command to determine the WWID:

     ```
     # /lib/udev/scsi_id -gud /dev/sda
     360030057024d0730239134810c0cb833
     ```

     `sda` is the local SCSI disk that we need to add it to the blacklist.

  2. Add the `WWID` to the blacklist stanza in `/etc/multipath.conf`:

     ```
     blacklist {
          wwid   360030057024d0730239134810c0cb833
          devnode "^(ram|raw|loop|fd|md|dm-|sr|scd|st)[0-9]*"
          devnode "^hd[a-z]"
          devnode "^cciss.*"
     }
     ```

You should always check your `/etc/multipath.conf` file for legacy settings, especially in the defaults section, that might be overriding the default settings.

The following table demonstrates the critical `multipathd` parameters for ONTAP LUNs and the required values. If a host is connected to LUNs from other vendors and any of these parameters are overridden, they will need to be corrected by later stanzas in the `multipath.conf` file that apply specifically to ONTAP LUNs. If this is not done, the ONTAP LUNs might not work as expected. These defaults should only be overridden in consultation with NetApp and/or an OS vendor and only when the impact is fully understood.

| Parameter | Setting |
|---|---|
| detect_prio | yes |
| dev_loss_tmo | "infinity" |
| failback | immediate |
| fast_io_fail_tmo | 5 |
| features | "3 queue_if_no_path pg_init_retries 50" |
| flush_on_last_del | "yes" |
| hardware_handler | "0" |
| no_path_retry | queue |
| path_checker | "tur" |
| path_grouping_policy | "group_by_prio" |
| path_selector | "round-robin 0" |
| polling_interval | 5 |
| prio | "ontap" |
| product | LUN.* |
| retain_attached_hw_handler | yes |
| rr_weight | "uniform" |
| user_friendly_names | no |
| vendor | NETAPP |

**Example**

The following example shows how to correct an overridden default. In this case, the `multipath.conf` file defines values for `path_checker` and `no_path_retry` that are not compatible with ONTAP LUNs. If they cannot be removed because of other SAN arrays still attached to the host, these parameters can be corrected specifically for ONTAP LUNs with a device stanza.

```
defaults {
   path_checker      readsector0
   no_path_retry      fail
}

devices {
   device {
      vendor         "NETAPP  "
      product         "LUN.*"
      no_path_retry     queue
      path_checker      tur
   }
}
```

#### KVM settings

You can use the recommended settings to configure Kernel-based Virtual Machine (KVM) as well. There are no changes required to configure KVM as the LUN is mapped to the hypervisor.

### Known issues

There are no known issues for the RHEL 6.8 with ONTAP release.

## Use Red Hat Enterprise Linux 6.7 with ONTAP

You can use the ONTAP SAN host configuration settings to configure Red Hat Enterprise Linux 6.7 with ONTAP as the target.

### Install the Linux Unified Host Utilities

The NetApp Linux Unified Host Utilities software package is available on the NetApp Support Site in a 32-bit and 64-bit .rpm file. If you do not know which file is right for your configuration, use the NetApp Interoperability Matrix Tool to verify which one you need.

NetApp strongly recommends installing the Linux Unified Host Utilities, but it is not mandatory. The utilities do not change any settings on your Linux host. The utilities improve management and assist NetApp customer support in gathering information about your configuration.

#### What you'll need

If you have a version of Linux Unified Host Utilities currently installed you should upgrade it or, you should remove it and use the following steps to install the latest version.

1. Download the 32-bit or 64-bit Linux Unified Host Utilities software package from the NetApp Support Site Site to your host.
2. Use the following command to install the software package:

   ```
   rpm -ivh netapp_linux_unified_host_utilities-7-1.x86_64
   ```

> You can use the configuration settings provided in this document to configure cloud clients connected to Cloud Volumes ONTAP and Amazon FSx for ONTAP.

## SAN Toolkit

The toolkit is installed automatically when you install the NetApp Host Utilities package. This kit provides the `sanlun` utility, which helps you manage LUNs and HBAs. The `sanlun` command returns information about the LUNs mapped to your host, multipathing, and information necessary to create initiator groups.

### Example

In the following example, the `sanlun lun show` command returns LUN information.

```
# sanlun lun show all
```

Example output:

```
controller(7mode/E-Series)/            device     host               lun
vserver(cDOT/FlashRay)   lun-pathname  filename   adapter  protocol  size
Product
------------------------------------------------------------------------
data_vserver          /vol/vol1/lun1   /dev/sdb   host16   FCP
120.0g  cDOT
data_vserver          /vol/vol1/lun1   /dev/sdc   host15   FCP
120.0g  cDOT
data_vserver          /vol/vol2/lun2   /dev/sdd   host16   FCP
120.0g  cDOT
data_vserver          /vol/vol2/lun2   /dev/sde   host15   FCP
120.0g  cDOT
```

## SAN Booting

### What you'll need

If you decide to use SAN booting, it must be supported by your configuration. You can use the NetApp Interoperability Matrix Tool to verify that your OS, HBA, HBA firmware and the HBA boot BIOS, and ONTAP version are supported.

### Steps

1. Map the SAN boot LUN to the host.
2. Verify that multiple paths are available.

   > Multiple paths become available after the host OS is up and running on the paths.

3. Enable SAN booting in the server BIOS for the ports to which the SAN boot LUN is mapped.

   For information on how to enable the HBA BIOS, see your vendor-specific documentation.

4. Reboot the host to verify that the boot is successful.

## Multipathing

For Red Hat Enterprise Linux (RHEL) 6.7 the /etc/multipath.conf file must exist, but you do not need to make specific changes to the file. RHEL 6.7 is compiled with all settings required to recognize and correctly manage ONTAP LUNs.
To Enable ALUA Handler, perform the following steps:

**Steps**

1. Create a backup of the initrd-image.
2. Append the following parameter value to the kernel for ALUA and non-ALUA to work:
   `rdloaddriver=scsi_dh_alua`

   **Example**

   ```
   kernel /vmlinuz-2.6.32-358.6.1.el6.x86_64 ro root=/dev/mapper/
   vg_ibmx355021082-lv_root rd_NO_LUKS rd_LVM_LV=vg_ibmx355021082/ lv_root
   LANG=en_US.UTF-8 rd_LVM_LV=vg_ibmx355021082/lv_swap rd_NO_MD
   SYSFONT=latarcyrheb-sun16 crashkernel=auto KEYBOARDTYPE=pc KEYTABLE=us
   rd_NO_DM rhgb quiet rdloaddriver=scsi_dh_alua
   ```

3. Use the `mkinitrd` command to recreate the initrd-image.
   RHEL 6x and later versions use either:
   The command: `mkinitrd -f /boot/ initrd-"uname -r".img uname -r`
   Or
   The command: `dracut -f`
4. Reboot the host.
5. Verify the output of the `cat /proc/cmdline` command to ensure that the setting is complete.

You can use the `multipath -ll` command to verify the settings for your ONTAP LUNs.
The following sections provide sample multipath output for a LUN mapped to ASA and non-ASA personas.

### All SAN Array configurations

In All SAN Array (ASA) configurations, all paths to a given LUN are active and optimized. This improves performance by serving I/O operations through all paths at the same time.

**Example**

The following example displays the correct output for an ONTAP LUN:

```
# multipath -ll
3600a0980383034466b2b4a3775474859 dm-3 NETAPP,LUN C-Mode
size=20G features='4 queue_if_no_path pg_init_retries 50
retain_attached_hw_handle' hwhandler='1 alua' wp=rw
|-+- policy='round-robin 0' prio=50 status=active
|- 1:0:8:1  sdb 8:16 active ready running
|- 2:0:8:1  sdd 8:48 active ready running
|- 1:0:9:1  sdc 8:32 active ready running
|- 2:0:9:1  sde 8:64 active ready running
```

Do not use an excessive number of paths to a single LUN. No more than four paths should be required. More than eight paths might cause path issues during storage failures.

#### Non-ASA configurations

For non-ASA configurations, there should be two groups of paths with different priorities. The paths with the higher priorities are Active/Optimized, meaning they are serviced by the controller where the aggregate is located. The paths with the lower priorities are active but are non-optimized because they are served from a different controller. The non-optimized paths are only used when no optimized paths are available.

**Example**

The following example displays the correct output for an ONTAP LUN with two Active/Optimized paths and two Active/Non-Optimized paths:

```
# multipath -ll
3600a0980383034466b2b4a3775474859 dm-3 NETAPP,LUN C-Mode
size=20G features='4 queue_if_no_path pg_init_retries 50
retain_attached_hw_handle' hwhandler='1 alua' wp=rw
|-+- policy='round-robin 0' prio=50 status=active
| |- 1:0:8:1  sdb 8:16 active ready running
| `- 2:0:8:1  sdd 8:48 active ready running
`-+- policy='round-robin 0' prio=10 status=enabled
  |- 1:0:9:1  sdc 8:32 active ready running
  `- 2:0:9:1  sde 8:64 active ready running
```

Do not use an excessive number of paths to a single LUN. No more than four paths should be required. More than eight paths might cause path issues during storage failures.

## Recommended Settings

The RHEL 6.7 OS is compiled to recognize ONTAP LUNs and automatically set all configuration parameters correctly for both ASA and non-ASA configuartion.

The `multipath.conf` file must exist for the multipath daemon to start, but you can create an empty, zero-byte file by using the following command:

```
touch /etc/multipath.conf
```

The first time you create this file, you might need to enable and start the multipath services:

```
# systemctl enable multipathd
# systemctl start multipathd
```

- There is no requirement to add anything directly to the `multipath.conf` file, unless you have devices that you do not want to be managed by multipath or you have existing settings that override defaults.
- To exclude unwanted devices, add the following syntax to the `multipath.conf` file .

  ```
  blacklist {
          wwid <DevId>
          devnode "^(ram|raw|loop|fd|md|dm-|sr|scd|st)[0-9]*"
          devnode "^hd[a-z]"
          devnode "^cciss.*"
  }
  ```

  Replace the `<DevId>` with the `WWID` string of the device you want to exclude.

  **Example**

  In this example, we are going to determine the WWID of a device and add to the `multipath.conf` file.

  **Steps**

  1. Run the following command to determine the WWID:

     ```
     # /lib/udev/scsi_id -gud /dev/sda
     360030057024d0730239134810c0cb833
     ```

     `sda` is the local SCSI disk that we need to add it to the blacklist.

  2. Add the `WWID` to the blacklist stanza in `/etc/multipath.conf`:

     ```
     blacklist {
          wwid   360030057024d0730239134810c0cb833
          devnode "^(ram|raw|loop|fd|md|dm-|sr|scd|st)[0-9]*"
          devnode "^hd[a-z]"
          devnode "^cciss.*"
     }
     ```

You should always check your `/etc/multipath.conf` file for legacy settings, especially in the defaults section, that might be overriding the default settings.

The following table demonstrates the critical `multipathd` parameters for ONTAP LUNs and the required values. If a host is connected to LUNs from other vendors and any of these parameters are overridden, they will need to be corrected by later stanzas in the `multipath.conf` file that apply specifically to ONTAP LUNs. If this is not done, the ONTAP LUNs might not work as expected. These defaults should only be overridden in consultation with NetApp and/or an OS vendor and only when the impact is fully understood.

| Parameter | Setting |
|---|---|
| detect_prio | yes |
| dev_loss_tmo | "infinity" |
| failback | immediate |
| fast_io_fail_tmo | 5 |
| features | "3 queue_if_no_path pg_init_retries 50" |
| flush_on_last_del | "yes" |
| hardware_handler | "0" |
| no_path_retry | queue |
| path_checker | "tur" |
| path_grouping_policy | "group_by_prio" |
| path_selector | "round-robin 0" |
| polling_interval | 5 |
| prio | "ontap" |
| product | LUN.* |
| retain_attached_hw_handler | yes |
| rr_weight | "uniform" |
| user_friendly_names | no |
| vendor | NETAPP |

**Example**

The following example shows how to correct an overridden default. In this case, the `multipath.conf` file defines values for `path_checker` and `no_path_retry` that are not compatible with ONTAP LUNs. If they cannot be removed because of other SAN arrays still attached to the host, these parameters can be corrected specifically for ONTAP LUNs with a device stanza.

```
defaults {
   path_checker      readsector0
   no_path_retry      fail
}

devices {
   device {
      vendor         "NETAPP  "
      product         "LUN.*"
      no_path_retry     queue
      path_checker      tur
   }
}
```

#### KVM settings

You can use the recommended settings to configure Kernel-based Virtual Machine (KVM) as well. There are no changes required to configure KVM as the LUN is mapped to the hypervisor.

### Known issues

There are no known issues for the RHEL 6.7 with ONTAP release.

## Use Red Hat Enterprise Linux 6.6 with ONTAP

You can use the ONTAP SAN host configuration settings to configure Red Hat Enterprise Linux 6.6 with ONTAP as the target.

### Install the Linux Unified Host Utilities

The NetApp Linux Unified Host Utilities software package is available on the NetApp Support Site in a 32-bit and 64-bit .rpm file. If you do not know which file is right for your configuration, use the NetApp Interoperability Matrix Tool to verify which one you need.

NetApp strongly recommends installing the Linux Unified Host Utilities, but it is not mandatory. The utilities do not change any settings on your Linux host. The utilities improve management and assist NetApp customer support in gathering information about your configuration.

#### What you'll need

If you have a version of Linux Unified Host Utilities currently installed you should upgrade it or, you should remove it and use the following steps to install the latest version.

1. Download the 32-bit or 64-bit Linux Unified Host Utilities software package from the NetApp Support Site Site to your host.
2. Use the following command to install the software package:

   ```
   rpm -ivh netapp_linux_unified_host_utilities-7-1.x86_64
   ```

> You can use the configuration settings provided in this document to configure cloud clients connected to Cloud Volumes ONTAP and Amazon FSx for ONTAP.

## SAN Toolkit

The toolkit is installed automatically when you install the NetApp Host Utilities package. This kit provides the `sanlun` utility, which helps you manage LUNs and HBAs. The `sanlun` command returns information about the LUNs mapped to your host, multipathing, and information necessary to create initiator groups.

### Example

In the following example, the `sanlun lun show` command returns LUN information.

```
# sanlun lun show all
```

Example output:

```
controller(7mode/E-Series)/            device     host               lun
vserver(cDOT/FlashRay)   lun-pathname  filename   adapter  protocol  size
Product
------------------------------------------------------------------------
data_vserver          /vol/vol1/lun1   /dev/sdb   host16   FCP
120.0g  cDOT
data_vserver          /vol/vol1/lun1   /dev/sdc   host15   FCP
120.0g  cDOT
data_vserver          /vol/vol2/lun2   /dev/sdd   host16   FCP
120.0g  cDOT
data_vserver          /vol/vol2/lun2   /dev/sde   host15   FCP
120.0g  cDOT
```

## SAN Booting

### What you'll need

If you decide to use SAN booting, it must be supported by your configuration. You can use the NetApp Interoperability Matrix Tool to verify that your OS, HBA, HBA firmware and the HBA boot BIOS, and ONTAP version are supported.

### Steps

1. Map the SAN boot LUN to the host.
2. Verify that multiple paths are available.

   > Multiple paths become available after the host OS is up and running on the paths.

3. Enable SAN booting in the server BIOS for the ports to which the SAN boot LUN is mapped.

   For information on how to enable the HBA BIOS, see your vendor-specific documentation.

4. Reboot the host to verify that the boot is successful.

## Multipathing

For Red Hat Enterprise Linux (RHEL) 6.6 the /etc/multipath.conf file must exist, but you do not need to make specific changes to the file. RHEL 6.6 is compiled with all settings required to recognize and correctly manage ONTAP LUNs.
To Enable ALUA Handler, perform the following steps:

**Steps**

1. Create a backup of the initrd-image.
2. Append the following parameter value to the kernel for ALUA and non-ALUA to work:
   `rdloaddriver=scsi_dh_alua`

   **Example**

   ```
   kernel /vmlinuz-2.6.32-358.6.1.el6.x86_64 ro root=/dev/mapper/
   vg_ibmx355021082-lv_root rd_NO_LUKS rd_LVM_LV=vg_ibmx355021082/ lv_root
   LANG=en_US.UTF-8 rd_LVM_LV=vg_ibmx355021082/lv_swap rd_NO_MD
   SYSFONT=latarcyrheb-sun16 crashkernel=auto KEYBOARDTYPE=pc KEYTABLE=us
   rd_NO_DM rhgb quiet rdloaddriver=scsi_dh_alua
   ```

3. Use the `mkinitrd` command to recreate the initrd-image.
   RHEL 6x and later versions use either:
   The command: `mkinitrd -f /boot/ initrd-"uname -r".img uname -r`
   Or
   The command: `dracut -f`
4. Reboot the host.
5. Verify the output of the `cat /proc/cmdline` command to ensure that the setting is complete.

You can use the `multipath -ll` command to verify the settings for your ONTAP LUNs.
The following sections provide sample multipath output for a LUN mapped to ASA and non-ASA personas.

### All SAN Array configurations

In All SAN Array (ASA) configurations, all paths to a given LUN are active and optimized. This improves performance by serving I/O operations through all paths at the same time.

**Example**

The following example displays the correct output for an ONTAP LUN:

```
# multipath -ll
3600a0980383034466b2b4a3775474859 dm-3 NETAPP,LUN C-Mode
size=20G features='4 queue_if_no_path pg_init_retries 50
retain_attached_hw_handle' hwhandler='1 alua' wp=rw
|-+- policy='round-robin 0' prio=50 status=active
|- 1:0:8:1  sdb 8:16 active ready running
|- 2:0:8:1  sdd 8:48 active ready running
|- 1:0:9:1  sdc 8:32 active ready running
|- 2:0:9:1  sde 8:64 active ready running
```

Do not use an excessive number of paths to a single LUN. No more than four paths should be required. More than eight paths might cause path issues during storage failures.

#### Non-ASA configurations

For non-ASA configurations, there should be two groups of paths with different priorities. The paths with the higher priorities are Active/Optimized, meaning they are serviced by the controller where the aggregate is located. The paths with the lower priorities are active but are non-optimized because they are served from a different controller. The non-optimized paths are only used when no optimized paths are available.

#### Example

The following example displays the correct output for an ONTAP LUN with two Active/Optimized paths and two Active/Non-Optimized paths:

```
# multipath -ll
3600a0980383034466b2b4a3775474859 dm-3 NETAPP,LUN C-Mode
size=20G features='4 queue_if_no_path pg_init_retries 50
retain_attached_hw_handle' hwhandler='1 alua' wp=rw
|-+- policy='round-robin 0' prio=50 status=active
| |- 1:0:8:1  sdb 8:16 active ready running
| `- 2:0:8:1  sdd 8:48 active ready running
`-+- policy='round-robin 0' prio=10 status=enabled
  |- 1:0:9:1  sdc 8:32 active ready running
  `- 2:0:9:1  sde 8:64 active ready running
```

Do not use an excessive number of paths to a single LUN. No more than four paths should be required. More than eight paths might cause path issues during storage failures.

## Recommended Settings

The RHEL 6.6 OS is compiled to recognize ONTAP LUNs and automatically set all configuration parameters correctly for both ASA and non-ASA configuration.

The `multipath.conf` file must exist for the multipath daemon to start, but you can create an empty, zero-byte file by using the following command:

```
touch /etc/multipath.conf
```

The first time you create this file, you might need to enable and start the multipath services:

```
# systemctl enable multipathd
# systemctl start multipathd
```

- There is no requirement to add anything directly to the `multipath.conf` file, unless you have devices that you do not want to be managed by multipath or you have existing settings that override defaults.
- To exclude unwanted devices, add the following syntax to the `multipath.conf` file .

  ```
  blacklist {
          wwid <DevId>
          devnode "^(ram|raw|loop|fd|md|dm-|sr|scd|st)[0-9]*"
          devnode "^hd[a-z]"
          devnode "^cciss.*"
  }
  ```

  Replace the `<DevId>` with the `WWID` string of the device you want to exclude.

  **Example**

  In this example, we are going to determine the WWID of a device and add to the `multipath.conf` file.

  **Steps**

  1. Run the following command to determine the WWID:

     ```
     # /lib/udev/scsi_id -gud /dev/sda
     360030057024d0730239134810c0cb833
     ```

     `sda` is the local SCSI disk that we need to add it to the blacklist.

  2. Add the `WWID` to the blacklist stanza in `/etc/multipath.conf`:

     ```
     blacklist {
          wwid   360030057024d0730239134810c0cb833
          devnode "^(ram|raw|loop|fd|md|dm-|sr|scd|st)[0-9]*"
          devnode "^hd[a-z]"
          devnode "^cciss.*"
     }
     ```

You should always check your `/etc/multipath.conf` file for legacy settings, especially in the defaults section, that might be overriding the default settings.

The following table demonstrates the critical `multipathd` parameters for ONTAP LUNs and the required values. If a host is connected to LUNs from other vendors and any of these parameters are overridden, they will need to be corrected by later stanzas in the `multipath.conf` file that apply specifically to ONTAP LUNs. If this is not done, the ONTAP LUNs might not work as expected. These defaults should only be overridden in consultation with NetApp and/or an OS vendor and only when the impact is fully understood.

| Parameter | Setting |
|---|---|
| detect_prio | yes |
| dev_loss_tmo | "infinity" |
| failback | immediate |
| fast_io_fail_tmo | 5 |
| features | "3 queue_if_no_path pg_init_retries 50" |
| flush_on_last_del | "yes" |
| hardware_handler | "0" |
| no_path_retry | queue |
| path_checker | "tur" |
| path_grouping_policy | "group_by_prio" |
| path_selector | "round-robin 0" |
| polling_interval | 5 |
| prio | "ontap" |
| product | LUN.* |
| retain_attached_hw_handler | yes |
| rr_weight | "uniform" |
| user_friendly_names | no |
| vendor | NETAPP |

**Example**

The following example shows how to correct an overridden default. In this case, the `multipath.conf` file defines values for `path_checker` and `no_path_retry` that are not compatible with ONTAP LUNs. If they cannot be removed because of other SAN arrays still attached to the host, these parameters can be corrected specifically for ONTAP LUNs with a device stanza.

```
defaults {
   path_checker       readsector0
   no_path_retry       fail
}

devices {
   device {
      vendor          "NETAPP  "
      product          "LUN.*"
      no_path_retry      queue
      path_checker       tur
   }
}
```

#### KVM settings

You can use the recommended settings to configure Kernel-based Virtual Machine (KVM) as well. There are no changes required to configure KVM as the LUN is mapped to the hypervisor.

## Known issues

The RHEL 6.6 with ONTAP release has the following known issues:

| NetApp Bug ID | Title | Description | Bugzilla ID |
|---|---|---|---|
| 863878 | Kernel crash occurs with RHEL 6U6 host during storage failures | You might observe kernel crash on RHEL 6U6 host during storage/fabric. | 1158363 |
| 1076584 | IO stall up to 300 sec seen with QLogic 16G FC (QLE2672) host during storage failures in RHEL 6U4 | You might observe an IO stall up to 300 sec on QLogic 16G FC (QLE2672) host during storage/fabric failures. | 1135962 |
| 795684 | RHEL6 U5 multipathd incorrectly group multipath maps during MoD and storage failover fault operations | You might observe an incorrect path grouping on LUNs during LUN move on demand operation along with storage faults. During LUN move operation multipath path priorities will change and multipath is unable to reloads the device table due to device failure caused by storage fault. This leads to incorrect path grouping. | 1151020 |

# Use Red Hat Enterprise Linux 6.5 with ONTAP

You can use the ONTAP SAN host configuration settings to configure Red Hat Enterprise Linux 6.5 with ONTAP as the target.

## Install the Linux Unified Host Utilities

The NetApp Linux Unified Host Utilities software package is available on the NetApp Support Site in a 32-bit and 64-bit .rpm file. If you do not know which file is right for your configuration, use the NetApp Interoperability Matrix Tool to verify which one you need.

NetApp strongly recommends installing the Linux Unified Host Utilities, but it is not mandatory. The utilities do not change any settings on your Linux host. The utilities improve management and assist NetApp customer support in gathering information about your configuration.

#### What you'll need

If you have a version of Linux Unified Host Utilities currently installed you should upgrade it or, you should remove it and use the following steps to install the latest version.

1. Download the 32-bit or 64-bit Linux Unified Host Utilities software package from the NetApp Support Site Site to your host.
2. Use the following command to install the software package:

   ```
   rpm -ivh netapp_linux_unified_host_utilities-7-1.x86_64
   ```

> You can use the configuration settings provided in this document to configure cloud clients connected to Cloud Volumes ONTAP and Amazon FSx for ONTAP.

## SAN Toolkit

The toolkit is installed automatically when you install the NetApp Host Utilities package. This kit provides the `sanlun` utility, which helps you manage LUNs and HBAs. The `sanlun` command returns information about the LUNs mapped to your host, multipathing, and information necessary to create initiator groups.

#### Example

In the following example, the `sanlun lun show` command returns LUN information.

```
# sanlun lun show all
```

Example output:

```
controller(7mode/E-Series)/            device     host                lun
vserver(cDOT/FlashRay)   lun-pathname  filename   adapter  protocol  size
Product
-------------------------------------------------------------------------
data_vserver          /vol/vol1/lun1   /dev/sdb   host16   FCP
120.0g  cDOT
data_vserver          /vol/vol1/lun1   /dev/sdc   host15   FCP
120.0g  cDOT
data_vserver          /vol/vol2/lun2   /dev/sdd   host16   FCP
120.0g  cDOT
data_vserver          /vol/vol2/lun2   /dev/sde   host15   FCP
120.0g  cDOT
```

## SAN Booting

### What you'll need

If you decide to use SAN booting, it must be supported by your configuration. You can use the NetApp Interoperability Matrix Tool to verify that your OS, HBA, HBA firmware and the HBA boot BIOS, and ONTAP version are supported.

### Steps

1. Map the SAN boot LUN to the host.
2. Verify that multiple paths are available.

   Multiple paths become available after the host OS is up and running on the paths.

3. Enable SAN booting in the server BIOS for the ports to which the SAN boot LUN is mapped.

   For information on how to enable the HBA BIOS, see your vendor-specific documentation.

4. Reboot the host to verify that the boot is successful.

## Multipathing

For Red Hat Enterprise Linux (RHEL) 6.5 the /etc/multipath.conf file must exist, but you do not need to make specific changes to the file. RHEL 6.5 is compiled with all settings required to recognize and correctly manage ONTAP LUNs.
To Enable ALUA Handler, perform the following steps:

### Steps

1. Create a backup of the initrd-image.
2. Append the following parameter value to the kernel for ALUA and non-ALUA to work:
   `rdloaddriver=scsi_dh_alua`

**Example**

```
kernel /vmlinuz-2.6.32-358.6.1.el6.x86_64 ro root=/dev/mapper/
vg_ibmx355021082-lv_root rd_NO_LUKS rd_LVM_LV=vg_ibmx355021082/ lv_root
LANG=en_US.UTF-8 rd_LVM_LV=vg_ibmx355021082/lv_swap rd_NO_MD
SYSFONT=latarcyrheb-sun16 crashkernel=auto KEYBOARDTYPE=pc KEYTABLE=us
rd_NO_DM rhgb quiet rdloaddriver=scsi_dh_alua
```

3. Use the `mkinitrd` command to recreate the initrd-image.
   RHEL 6x and later versions use either:
   The command: `mkinitrd -f /boot/ initrd-"uname -r".img uname -r`
   Or
   The command: `dracut -f`
4. Reboot the host.
5. Verify the output of the `cat /proc/cmdline` command to ensure that the setting is complete.

You can use the `multipath -ll` command to verify the settings for your ONTAP LUNs.
The following sections provide sample multipath output for a LUN mapped to ASA and non-ASA personas.

### All SAN Array configurations

In All SAN Array (ASA) configurations, all paths to a given LUN are active and optimized. This improves performance by serving I/O operations through all paths at the same time.

**Example**

The following example displays the correct output for an ONTAP LUN:

```
# multipath -ll
3600a0980383034466b2b4a3775474859 dm-3 NETAPP,LUN C-Mode
size=20G features='4 queue_if_no_path pg_init_retries 50
retain_attached_hw_handle' hwhandler='1 alua' wp=rw
|-+- policy='round-robin 0' prio=50 status=active
|- 1:0:8:1  sdb 8:16 active ready running
|- 2:0:8:1  sdd 8:48 active ready running
|- 1:0:9:1  sdc 8:32 active ready running
|- 2:0:9:1  sde 8:64 active ready running
```

> Do not use an excessive number of paths to a single LUN. No more than four paths should be required. More than eight paths might cause path issues during storage failures.

### Non-ASA configurations

For non-ASA configurations, there should be two groups of paths with different priorities. The paths with the higher priorities are Active/Optimized, meaning they are serviced by the controller where the aggregate is located. The paths with the lower priorities are active but are non-optimized because they are served from a different controller. The non-optimized paths are only used when no optimized paths are available.

**Example**

The following example displays the correct output for an ONTAP LUN with two Active/Optimized paths and two Active/Non-Optimized paths:

```
# multipath -ll
3600a0980383034466b2b4a3775474859 dm-3 NETAPP,LUN C-Mode
size=20G features='4 queue_if_no_path pg_init_retries 50
retain_attached_hw_handle' hwhandler='1 alua' wp=rw
|-+- policy='round-robin 0' prio=50 status=active
| |- 1:0:8:1  sdb 8:16 active ready running
| `- 2:0:8:1  sdd 8:48 active ready running
`-+- policy='round-robin 0' prio=10 status=enabled
  |- 1:0:9:1  sdc 8:32 active ready running
  `- 2:0:9:1  sde 8:64 active ready running
```

Do not use an excessive number of paths to a single LUN. No more than four paths should be required. More than eight paths might cause path issues during storage failures.

## Recommended Settings

The RHEL 6.5 OS is compiled to recognize ONTAP LUNs and automatically set all configuration parameters correctly for both ASA and non-ASA configuration.

The `multipath.conf` file must exist for the multipath daemon to start, but you can create an empty, zero-byte file by using the following command:

```
touch /etc/multipath.conf
```

The first time you create this file, you might need to enable and start the multipath services:

```
# systemctl enable multipathd
# systemctl start multipathd
```

- There is no requirement to add anything directly to the `multipath.conf` file, unless you have devices that you do not want to be managed by multipath or you have existing settings that override defaults.
- To exclude unwanted devices, add the following syntax to the `multipath.conf` file .

  ```
  blacklist {
          wwid <DevId>
          devnode "^(ram|raw|loop|fd|md|dm-|sr|scd|st)[0-9]*"
          devnode "^hd[a-z]"
          devnode "^cciss.*"
  }
  ```

  Replace the `<DevId>` with the `WWID` string of the device you want to exclude.

**Example**

In this example, we are going to determine the WWID of a device and add to the `multipath.conf` file.

**Steps**

1. Run the following command to determine the WWID:

```
# /lib/udev/scsi_id -gud /dev/sda
360030057024d0730239134810c0cb833
```

   `sda` is the local SCSI disk that we need to add it to the blacklist.

2. Add the `WWID` to the blacklist stanza in `/etc/multipath.conf`:

```
blacklist {
     wwid   360030057024d0730239134810c0cb833
     devnode "^(ram|raw|loop|fd|md|dm-|sr|scd|st)[0-9]*"
     devnode "^hd[a-z]"
     devnode "^cciss.*"
}
```

You should always check your `/etc/multipath.conf` file for legacy settings, especially in the defaults section, that might be overriding the default settings.

The following table demonstrates the critical `multipathd` parameters for ONTAP LUNs and the required values. If a host is connected to LUNs from other vendors and any of these parameters are overridden, they will need to be corrected by later stanzas in the `multipath.conf` file that apply specifically to ONTAP LUNs. If this is not done, the ONTAP LUNs might not work as expected. These defaults should only be overridden in consultation with NetApp and/or an OS vendor and only when the impact is fully understood.

| Parameter | Setting |
|---|---|
| detect_prio | yes |
| dev_loss_tmo | "infinity" |
| failback | immediate |
| fast_io_fail_tmo | 5 |
| features | "3 queue_if_no_path pg_init_retries 50" |
| flush_on_last_del | "yes" |
| hardware_handler | "0" |
| no_path_retry | queue |
| path_checker | "tur" |
| path_grouping_policy | "group_by_prio" |
| path_selector | "round-robin 0" |

| Parameter | Setting |
| --- | --- |
| polling_interval | 5 |
| prio | "ontap" |
| product | LUN.* |
| retain_attached_hw_handler | yes |
| rr_weight | "uniform" |
| user_friendly_names | no |
| vendor | NETAPP |

#### Example

The following example shows how to correct an overridden default. In this case, the `multipath.conf` file defines values for `path_checker` and `no_path_retry` that are not compatible with ONTAP LUNs. If they cannot be removed because of other SAN arrays still attached to the host, these parameters can be corrected specifically for ONTAP LUNs with a device stanza.

```
defaults {
   path_checker      readsector0
   no_path_retry      fail
}

devices {
   device {
      vendor         "NETAPP  "
      product         "LUN.*"
      no_path_retry     queue
      path_checker      tur
   }
}
```

#### KVM settings

You can use the recommended settings to configure Kernel-based Virtual Machine (KVM) as well. There are no changes required to configure KVM as the LUN is mapped to the hypervisor.

## Known issues

The RHEL 6.5 with ONTAP release has the following known issues:

| NetApp Bug ID | Title | Description | Bugzilla ID |
| --- | --- | --- | --- |
| 760515 | Path failures or host hangs were observed in RHEL 6.5 8G Qlogic FC SAN host during storage failover operations | Path failures or host hangs were observed in RHEL 6.5 8G Qlogic FC SAN host during storage failover operations. | 1033136 |

| NetApp Bug ID | Title | Description | Bugzilla ID |
|---|---|---|---|
| 758271 | bnx2 firmware fails to load when booting with custom initrd (dracut -f) | Broadcom NetXtreme II Gigabit controller ports will not ping due to bnx2 firmware fails to load during boot with custom initrd. | 1007463 |
| 799394 | RHEL 6U5: Emulex 16G FC (LPe16002B-M6) host crash is seen during I/O with storage failover operations | 16G FC Emulex (LPe16002B-M6) host crash is seen during I/O with storage failover operations. | 1063699 |
| 786571 | QLogic FCoE host hangs/path failures observed in RHEL 6.5 during I/O with storage failover operations | QLogic FCoE (QLE8242) host hangs/path failures are observed in RHEL 6.5 during I/O with storage failover operations. In such scenarios, you might see the following message: "Mailbox cmd timeout occurred, cmd=0x54, mb[0]=0x54. Scheduling ISP abort" messages which leads to host hung/path failures. | 1068619 |
| 801580 | QLogic 16G FC host hangs or path failures observed in RHEL 6.5 during I/O with storage failover operations | The I/O delays of more than 600 seconds are observed with QLogic 16G FC host (QLE2672) during storage failover operations. In such scenarios, the following message is displayed: "Failed mbx[0]=54, mb[1]=0, mb[2]=76b9, mb[3]=5200, cmd=54" | 1068622 |

# Use Red Hat Enterprise Linux 6.4 with ONTAP

You can use the ONTAP SAN host configuration settings to configure Red Hat Enterprise Linux 6.4 with ONTAP as the target.

## Install the Linux Unified Host Utilities

The NetApp Linux Unified Host Utilities software package is available on the NetApp Support Site in a 32-bit and 64-bit .rpm file. If you do not know which file is right for your configuration, use the NetApp Interoperability Matrix Tool to verify which one you need.

NetApp strongly recommends installing the Linux Unified Host Utilities, but it is not mandatory. The utilities do not change any settings on your Linux host. The utilities improve management and assist NetApp customer

support in gathering information about your configuration.

**What you'll need**

If you have a version of Linux Unified Host Utilities currently installed you should upgrade it or, you should remove it and use the following steps to install the latest version.

1. Download the 32-bit or 64-bit Linux Unified Host Utilities software package from the NetApp Support Site Site to your host.
2. Use the following command to install the software package:

```
rpm -ivh netapp_linux_unified_host_utilities-7-1.x86_64
```

> You can use the configuration settings provided in this document to configure cloud clients connected to Cloud Volumes ONTAP and Amazon FSx for ONTAP.

## SAN Toolkit

The toolkit is installed automatically when you install the NetApp Host Utilities package. This kit provides the `sanlun` utility, which helps you manage LUNs and HBAs. The `sanlun` command returns information about the LUNs mapped to your host, multipathing, and information necessary to create initiator groups.

**Example**

In the following example, the `sanlun lun show` command returns LUN information.

```
# sanlun lun show all
```

Example output:

```
controller(7mode/E-Series)/            device     host               lun
vserver(cDOT/FlashRay)   lun-pathname  filename   adapter  protocol  size
Product
------------------------------------------------------------------------
data_vserver          /vol/vol1/lun1   /dev/sdb   host16   FCP
120.0g  cDOT
data_vserver          /vol/vol1/lun1   /dev/sdc   host15   FCP
120.0g  cDOT
data_vserver          /vol/vol2/lun2   /dev/sdd   host16   FCP
120.0g  cDOT
data_vserver          /vol/vol2/lun2   /dev/sde   host15   FCP
120.0g  cDOT
```

## SAN Booting

**What you'll need**

If you decide to use SAN booting, it must be supported by your configuration. You can use the NetApp Interoperability Matrix Tool to verify that your OS, HBA, HBA firmware and the HBA boot BIOS, and ONTAP

version are supported.

**Steps**

1. Map the SAN boot LUN to the host.
2. Verify that multiple paths are available.

   Multiple paths become available after the host OS is up and running on the paths.

3. Enable SAN booting in the server BIOS for the ports to which the SAN boot LUN is mapped.

   For information on how to enable the HBA BIOS, see your vendor-specific documentation.

4. Reboot the host to verify that the boot is successful.

## Multipathing

For Red Hat Enterprise Linux (RHEL) 6.4 the /etc/multipath.conf file must exist, but you do not need to make specific changes to the file. RHEL 6.4 is compiled with all settings required to recognize and correctly manage ONTAP LUNs.
To Enable ALUA Handler, perform the following steps:

**Steps**

1. Create a backup of the initrd-image.
2. Append the following parameter value to the kernel for ALUA and non-ALUA to work:
   `rdloaddriver=scsi_dh_alua`

   **Example**

   ```
   kernel /vmlinuz-2.6.32-358.6.1.el6.x86_64 ro root=/dev/mapper/
   vg_ibmx355021082-lv_root rd_NO_LUKS rd_LVM_LV=vg_ibmx355021082/ lv_root
   LANG=en_US.UTF-8 rd_LVM_LV=vg_ibmx355021082/lv_swap rd_NO_MD
   SYSFONT=latarcyrheb-sun16 crashkernel=auto KEYBOARDTYPE=pc KEYTABLE=us
   rd_NO_DM rhgb quiet rdloaddriver=scsi_dh_alua
   ```

3. Use the `mkinitrd` command to recreate the initrd-image.
   RHEL 6x and later versions use either:
   The command: `mkinitrd -f /boot/ initrd-"uname -r".img uname -r`
   Or
   The command: `dracut -f`
4. Reboot the host.
5. Verify the output of the `cat /proc/cmdline` command to ensure that the setting is complete.

You can use the `multipath -ll` command to verify the settings for your ONTAP LUNs.
The following sections provide sample multipath output for a LUN mapped to ASA and non-ASA personas.

### All SAN Array configurations

In All SAN Array (ASA) configurations, all paths to a given LUN are active and optimized. This improves performance by serving I/O operations through all paths at the same time.

#### Example

The following example displays the correct output for an ONTAP LUN:

```
# multipath -ll
3600a0980383034466b2b4a3775474859 dm-3 NETAPP,LUN C-Mode
size=20G features='4 queue_if_no_path pg_init_retries 50
retain_attached_hw_handle' hwhandler='1 alua' wp=rw
|-+- policy='round-robin 0' prio=50 status=active
|- 1:0:8:1  sdb 8:16 active ready running
|- 2:0:8:1  sdd 8:48 active ready running
|- 1:0:9:1  sdc 8:32 active ready running
|- 2:0:9:1  sde 8:64 active ready running
```

Do not use an excessive number of paths to a single LUN. No more than four paths should be required. More than eight paths might cause path issues during storage failures.

#### Non-ASA configurations

For non-ASA configurations, there should be two groups of paths with different priorities. The paths with the higher priorities are Active/Optimized, meaning they are serviced by the controller where the aggregate is located. The paths with the lower priorities are active but are non-optimized because they are served from a different controller. The non-optimized paths are only used when no optimized paths are available.

#### Example

The following example displays the correct output for an ONTAP LUN with two Active/Optimized paths and two Active/Non-Optimized paths:

```
# multipath -ll
3600a0980383034466b2b4a3775474859 dm-3 NETAPP,LUN C-Mode
size=20G features='4 queue_if_no_path pg_init_retries 50
retain_attached_hw_handle' hwhandler='1 alua' wp=rw
|-+- policy='round-robin 0' prio=50 status=active
| |- 1:0:8:1  sdb 8:16 active ready running
| `- 2:0:8:1  sdd 8:48 active ready running
`-+- policy='round-robin 0' prio=10 status=enabled
  |- 1:0:9:1  sdc 8:32 active ready running
  `- 2:0:9:1  sde 8:64 active ready running
```

Do not use an excessive number of paths to a single LUN. No more than four paths should be required. More than eight paths might cause path issues during storage failures.

## Recommended Settings

The RHEL 6.4 OS is compiled to recognize ONTAP LUNs and automatically set all configuration parameters correctly for both ASA and non-ASA configuration.

The `multipath.conf` file must exist for the multipath daemon to start, but you can create an empty, zero-byte file by using the following command:

```
touch /etc/multipath.conf
```

The first time you create this file, you might need to enable and start the multipath services:

```
# systemctl enable multipathd
# systemctl start multipathd
```

- There is no requirement to add anything directly to the `multipath.conf` file, unless you have devices that you do not want to be managed by multipath or you have existing settings that override defaults.
- To exclude unwanted devices, add the following syntax to the `multipath.conf` file .

  ```
  blacklist {
          wwid <DevId>
          devnode "^(ram|raw|loop|fd|md|dm-|sr|scd|st)[0-9]*"
          devnode "^hd[a-z]"
          devnode "^cciss.*"
  }
  ```

  Replace the `<DevId>` with the `WWID` string of the device you want to exclude.

  **Example**

  In this example, we are going to determine the WWID of a device and add to the `multipath.conf` file.

  **Steps**

  1. Run the following command to determine the WWID:

     ```
     # /lib/udev/scsi_id -gud /dev/sda
     360030057024d0730239134810c0cb833
     ```

     `sda` is the local SCSI disk that we need to add it to the blacklist.

  2. Add the `WWID` to the blacklist stanza in `/etc/multipath.conf`:

     ```
     blacklist {
          wwid   360030057024d0730239134810c0cb833
          devnode "^(ram|raw|loop|fd|md|dm-|sr|scd|st)[0-9]*"
          devnode "^hd[a-z]"
          devnode "^cciss.*"
     }
     ```

You should always check your `/etc/multipath.conf` file for legacy settings, especially in the defaults section, that might be overriding the default settings.

The following table demonstrates the critical `multipathd` parameters for ONTAP LUNs and the required values. If a host is connected to LUNs from other vendors and any of these parameters are overridden, they will need to be corrected by later stanzas in the `multipath.conf` file that apply specifically to ONTAP LUNs. If this is not done, the ONTAP LUNs might not work as expected. These defaults should only be overridden in consultation with NetApp and/or an OS vendor and only when the impact is fully understood.

| Parameter | Setting |
|---|---|
| detect_prio | yes |
| dev_loss_tmo | "infinity" |
| failback | immediate |
| fast_io_fail_tmo | 5 |
| features | "3 queue_if_no_path pg_init_retries 50" |
| flush_on_last_del | "yes" |
| hardware_handler | "0" |
| no_path_retry | queue |
| path_checker | "tur" |
| path_grouping_policy | "group_by_prio" |
| path_selector | "round-robin 0" |
| polling_interval | 5 |
| prio | "ontap" |
| product | LUN.* |
| retain_attached_hw_handler | yes |
| rr_weight | "uniform" |
| user_friendly_names | no |
| vendor | NETAPP |

**Example**

The following example shows how to correct an overridden default. In this case, the `multipath.conf` file defines values for `path_checker` and `no_path_retry` that are not compatible with ONTAP LUNs. If they cannot be removed because of other SAN arrays still attached to the host, these parameters can be corrected specifically for ONTAP LUNs with a device stanza.

```
defaults {
   path_checker      readsector0
   no_path_retry      fail
}

devices {
   device {
      vendor         "NETAPP  "
      product         "LUN.*"
      no_path_retry     queue
      path_checker      tur
   }
}
```

#### KVM settings

You can use the recommended settings to configure Kernel-based Virtual Machine (KVM) as well. There are no changes required to configure KVM as the LUN is mapped to the hypervisor.

## Known issues

The RHEL 6.4 with ONTAP release has the following known issues:

| NetApp Bug ID | Title | Description | Bugzilla ID |
| --- | --- | --- | --- |
| 673009 | Creating ext4 file system on LV striped across 15 or more discard-enabled, thinly provisioned multipath devices triggers "request botched" kernel errors | "Request blotched" kernel errors have been seen when users attempt to create an ext4 file system on discard-enabled, thinly provisioned multipath devices. As a result, creating the ext4 file system might take longer to complete and occasional disruption occurs. This issue has occurred only when users attempt to create the ext4 file system on a LV striped across 15 or more discard-enabled multipath devices on systems running Red Hat Enterprise Linux 6.x and Data ONTAP 8.1.3 and later operating in 7-Mode. The issue happens because the kernel erroneously attempts to merge discard requests, which is not supported on Red Hat Enterprise Linux 6.x at this time. When this issue occurs, multiple instances of the following message are written to syslog (/var/log/messages): kernel: blk: request botched. As a result, file system creation might take longer time to complete than expected. | 907844 |

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